

PESKINE AND SZPIRO'S DIMENSION INEQUALITY HOLDS OVER LOCAL RINGS OF DEPTH AT MOST ONE

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ABSTRACT. We prove that the general version—the one that includes the dimension of the tensor product—of Peskine and Szpiro's Dimension Inequality holds over noetherian local rings of depth at most one.

Let R be a commutative noetherian local ring. Serre [7] proved that if R is regular and M and N are finitely generated R -modules, then the inequality

$$\dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R N)$$

holds. It was conjectured by Peskine and Szpiro [6] that this inequality also would hold over a singular local ring, as long as one of the modules, say M , has finite projective dimension. In this paper, we will refer to this inequality as the *Dimension Inequality*, though some authors reserve that name for the special case where the R -module $M \otimes_R N$ has finite length and hence dimension 0.

An example by Ma [4, §2.3] shows that the Dimension Inequality does not hold in general. To be precise, Ma constructs a Cohen–Macaulay local ring of dimension 3 over which the inequality fails. In that light, one asks under what extra assumptions on the ring the inequality is valid. Our contribution is the following.

Theorem 1. *Let R be a commutative noetherian local ring and M and N be finitely generated R -modules. If $\mathrm{pd}_R M$ is finite and $\mathrm{depth} R \leq 1$ holds, then one has*

$$(1) \quad \dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R N).$$

Proof. By Peskine and Szpiro's Intersection Theorem, the inequality

$$(2) \quad \dim_R N \leq \mathrm{pd}_R M + \dim_R(M \otimes_R N)$$

holds. It follows from the Auslander–Buchsbaum Formula and the assumptions on M and R that $\mathrm{pd}_R M$ is 0 or 1. In the former case, M is free, so $\dim_R M = \dim R$ holds and thus (1) follows from (2) by adding this quantity to both side.

Now assume that $\mathrm{pd}_R M = 1$ holds and consider the minimal free resolution

$$0 \longrightarrow R^a \longrightarrow R^b \longrightarrow M \longrightarrow 0$$

where $a \leq b$ are positive integers. For every minimal prime ideal $\mathfrak{p}$ in R it induces an exact sequence

$$(3) \quad 0 \longrightarrow R_{\mathfrak{p}}^a \longrightarrow R_{\mathfrak{p}}^b \longrightarrow M_{\mathfrak{p}} \longrightarrow 0$$

of finite length $R_{\mathfrak{p}}$ -modules.

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If $a = b$ holds, then (3) yields $\text{length}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 0$ and thus $M_{\mathfrak{p}} = 0$ for every minimal prime ideal $\mathfrak{p}$ of R . It follows that $\dim_R M \leq \dim R - 1$ holds, and adding this to (2) yields:

$$\begin{aligned} \dim_R M + \dim_R N &\leq \dim R - 1 + \text{pd}_R M + \dim_R(M \otimes_R N) \\ &= \dim R + \dim_R(M \otimes_R N). \end{aligned}$$

If $a < b$ holds, then (3) yields

$$\text{length}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = (b - a) \text{length}_{R_{\mathfrak{p}}}(R_{\mathfrak{p}}) > 0$$

and, therefore, $M_{\mathfrak{p}} \neq 0$ for every minimal prime ideal $\mathfrak{p}$ in R . Thus, $\text{Supp}_R M$ is all of $\text{Spec } R$, whence one has

$$\text{Supp}_R(M \otimes_R N) = \text{Supp}_R M \cap \text{Supp}_R N = \text{Supp}_R N.$$

In particular, the equalities $\dim_R M = \dim R$ and $\dim_R N = \dim_R(M \otimes_R N)$ hold, whence equality holds in (1). $\square$

Remark 2. The assumption on the projective dimension of M in Theorem 1 can not be dropped; see [2, Exa. 18.5.4]. Also, the proof shows that, the assumption $\text{depth } R \leq 1$ can be replaced by the weaker assumption $\text{pd}_R M \leq 1$.

Theorem 1 extends to complexes as follows:

Corollary 3. *Let R be a commutative noetherian local ring and M and N be R -complexes with finitely generated homology modules and $H_i(M) = 0 = H_i(N)$ for $i \ll 0$. If $\text{pd}_R(H_i(M))$ is finite for all $i \in \mathbb{Z}$ and $\text{depth } R \leq 1$ holds, then one has*

$$\dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R^{\mathbf{L}} N).$$

Proof. The proof of [2, Thm. 20.1.34] applies verbatim: It is a standard reduction to the module case and the assumption on M ensures that Theorem 1 applies. $\square$

Remark 4. If the R -complex M in Corollary 3 has bounded homology, then the assumption on the homology modules implies that M has finite projective dimension; see for example Aihara et. al [1, Thm. 4.1]

This project arose from the question, does the Dimension Inequality hold over fiber product rings? As the next example shows, the answer is affirmative for a particular class of fiber products. These fiber product rings are precisely the local rings with decomposable maximal ideal; see Ogoma [5, Lem. 3.1].

Example 5. Let $(S, \mathfrak{s}, k)$ and $(T, \mathfrak{t}, k)$ be commutative noetherian local rings such that $S \neq k \neq T$ and let $R = S \times_k T$ be the fiber product of S and T over k , i.e.

$$R = \{(s, t) \in S \times T \mid \pi_S(s) = \pi_T(t)\}$$

where $S \xrightarrow{\pi_S} k \xleftarrow{\pi_T} T$ are the natural surjections. It is a commutative noetherian local ring with $\mathfrak{m} = \mathfrak{s} \oplus \mathfrak{t}$ and residue field k . By work of Lescot [3] one has

$$\text{depth } R = \min\{\text{depth } S, \text{depth } T, 1\}.$$

In particular, $\text{depth } R \leq 1$ holds and, therefore, Theorem 1 applies.

Remark 6. The ring considered in [2, Exa. 18.5.14]—already mentioned in Remark 2—is the fiber product $k[[x]] \times_k k[[y]]$. Thus, the assumption on the projective dimension of M can not be dropped for fiber products.

The local ring in Ma's [4, §2.3] counterexample to the Dimension Inequality is also a fiber product but, of course, different from those considered above: Ma's ring $A \times_C B$ is glued over a ring C of dimension 2 rather than over the residue field.

Here is a local ring that is covered by Theorem 1 and not a fiber product:

Example 7. The ring $R = k[[x, y]]/(y^2 - x^3)$ is a Cohen–Macaulay local domain (hence not a fiber product) with $\dim R = 1$, so Theorem 1 applies to R . Set $M = R/(x)$ and $N = \mathfrak{m}$. As x is not a zerodivisor, one has $\mathrm{pd}_R M = 1$ and $\dim_R M = 0 = \dim_R(M \otimes_R N)$. Since R is a domain, $\dim_R N = \dim R = 1$ holds. Thus equality holds in the Dimension Inequality (1) as well as in the inequality (2).

In closing we note that the proof of [4, Prop. B.5(1)] adapts to show that also the general version of the Dimension Inequality holds for rings of dimension at most 2.

Proposition 8. *Let R be a commutative noetherian local ring and M and N be finitely generated R -modules. If $\mathrm{pd}_R M$ is finite and $\dim R \leq 2$ holds, then one has*

$$\dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R N) .$$

Proof. First note that by Theorem 1 we can assume that $\dim R = 2$ holds.

If $\dim_R M = 0$ holds, then the asserted inequality holds trivially. If $\dim_R M = 1$ holds, then there is, as in the proof of [4, Prop. B.5], a non-zero-divisor $z \in R$ with $V(z) \subseteq \mathrm{Supp}_R M$ and, therefore,

$$\mathrm{Supp}_R(N/zN) = \mathrm{Supp}_R N \cap V(z) \subseteq \mathrm{Supp}_R N \cap \mathrm{Supp}_R M = \mathrm{Supp}_R(M \otimes_R N) .$$

Thus, $\dim_R(N/zN) \leq \dim_R(M \otimes_R N)$ holds, whence one has

$$\dim_R N \leq 1 + \dim_R(N/zN) \leq 1 + \dim_R(M \otimes_R N) .$$

If $\dim_R M = 2$ holds, then $\mathrm{Supp}_R M = \mathrm{Spec} R$, see again the proof of [4, Prop. B.5], and thus $\mathrm{Supp}_R(M \otimes_R N) = \mathrm{Supp}_R N$ holds. In particular, one has $\dim_R N = \dim_R(M \otimes_R N)$, and the asserted inequality follows; actually equality holds. $\square$

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